5.3 Quiz: Maclaurin Series and Kinematics — Markscheme

Calculator allowed. Total: 53 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. Where a part asks for an exact form or a series, a GDC decimal earns the method marks shown but not the final A1.

1. [Maximum mark: 6]

(a) $e^{-2x} = 1 + (-2x) + \dfrac{(-2x)^2}{2!} + \dfrac{(-2x)^3}{3!}$ **M1**

$= 1 - 2x + 2x^2 - \dfrac{4}{3}x^3$ **A1**

M1 for substituting $-2x$ into the series for e^x. Wrong method: leaving out the factorials ($1 - 2x + 4x^2 - 8x^3$): M1A0.

(b) $(1 + 3x)\left(1 - 2x + 2x^2 - \frac{4}{3}x^3\right)$ **M1**

$= 1 - 2x + 2x^2 - \frac{4}{3}x^3 + 3x - 6x^2 + 6x^3$ **A1**

$= 1 + x - 4x^2 + \dfrac{14}{3}x^3$ **A1**

M1 for multiplying their series by $1 + 3x$. A1 for all products up to x^3 (no others needed). FT from (a). Accept a derivative route ($f(0)$, $f'(0)$, ...) for full marks.

(c) $f''(0) = 2! \times (-4) = -8$ **A1**

Wrong method: the coefficient -4 itself (forgetting the $2!$): A0. FT from (b).

2. [Maximum mark: 7]

(a) $a = \dfrac{\mathrm{d}v}{\mathrm{d}t} = 2t - 7$ **M1**

$a(4) = 1 \text{ m s}^{-2}$ **A1**

Wrong method: $v(4) = -2$ is the velocity: M0A0.

(b) $t^2 - 7t + 10 = 0, \quad (t-2)(t-5) = 0$ **M1**

$t = 2, \ t = 5$ **A1**

A1 for both values.

(c) distance $= \displaystyle\int_0^6 \left|t^2 - 7t + 10\right| \mathrm{d}t$ **M1**

$= \dfrac{26}{3} + \dfrac{9}{2} + \dfrac{11}{6}$ **(A1)**

$= 15 \text{ m}$ **A1**

M1 for the modulus integral, or for splitting at $t = 2$ and $t = 5$ and adding the sizes. The GDC value of the modulus integral is acceptable with the integral shown. Wrong methods: the displacement $\int_0^6 v \, \mathrm{d}t = 6$: M0A0A0; splitting only at $t = 2$ ($\frac{26}{3} + \frac{8}{3} = \frac{34}{3}$): M1A0A0.

3. [Maximum mark: 13]

(a) $f'(x) = \dfrac{\cos x}{1+\sin x}$ **A1**

$f''(x) = \dfrac{-\sin x\,(1+\sin x) - \cos x \cos x}{(1+\sin x)^2}$ **M1**

$= \dfrac{-\left(\sin x + \sin^2 x + \cos^2 x\right)}{(1+\sin x)^2} = \dfrac{-(1+\sin x)}{(1+\sin x)^2}$ **A1**

$= -\dfrac{1}{1+\sin x}$ **AG**

A1 for the chain rule result. M1 for the quotient (or product) rule. A1 for using $\sin^2 x + \cos^2 x = 1$*; the AG line alone earns A0.*

(b) $f''(x) = -(1+\sin x)^{-1}$, so $f'''(x) = (1+\sin x)^{-2}\cos x$ **M1**

$f'''(x) = \dfrac{\cos x}{(1+\sin x)^2}$ **A1**

(c) $f(0) = 0,\ \ f'(0) = 1,\ \ f''(0) = -1,\ \ f'''(0) = 1$ **A1**

$f(x) \approx 0 + x - \dfrac{x^2}{2!} + \dfrac{x^3}{3!}$ **M1**

$f(x) \approx x - \dfrac{x^2}{2} + \dfrac{x^3}{6}$ **A1**

A1 for all four values. M1 for the Maclaurin formula with their values. Wrong method: dividing by n *instead of* $n!$ *gives* $\frac{x^3}{3}$*: M0 for that term, A0. FT from (b).*

(d) $\dfrac{\left(x - \frac{x^2}{2} + \frac{x^3}{6}\right) - x}{x^2} = -\dfrac{1}{2} + \dfrac{x}{6}$ **M1**

$\to -\dfrac{1}{2}$ **A1**

The question says to use part (c): l'Hôpital's rule applied twice with the correct answer earns M0A1. FT from (c).

(e) $\displaystyle\int_0^{0.4} \left(x - \frac{x^2}{2} + \frac{x^3}{6}\right) \mathrm{d}x = \left[\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}\right]_0^{0.4}$ **M1**

$= 0.0704 \quad$ (exactly $\frac{44}{625}$) **A1**

$\text{percentage error} = \dfrac{|0.0704 - 0.0702534\ldots|}{0.0702534\ldots} \times 100 = 0.209\% \quad (0.208666\ldots)$ **A1**

The GDC value of the integral is $0.0702534\ldots$*. Accept any value consistent with the candidate's own series. Wrong method: dividing by the approximation instead of the true value gives* 0.208%*: A0 for the percentage error. FT from (c).*

4. [Maximum mark: 14]

(a) $\dfrac{\mathrm{d}v}{\mathrm{d}t} = -0.02v^2, \quad \displaystyle\int v^{-2}\,\mathrm{d}v = \int -0.02\,\mathrm{d}t$ **M1**

$-\dfrac{1}{v} = -0.02t + C$ **A1**

$t = 0,\ v = 30: \quad C = -\dfrac{1}{30}$ **M1**

$\dfrac{1}{v} = 0.02t + \dfrac{1}{30} = \dfrac{0.6t+1}{30}$ **A1**

$v = \dfrac{30}{1+0.6t}$ **AG**

M1 for separating the variables. M1 for using the initial condition. The final A1 needs a line that leads to the given form. Verifying the given answer by differentiation earns no marks: the question says "by solving".

(b) $\dfrac{30}{1+0.6t} = 12, \quad 1+0.6t = 2.5$ **M1**

$t = 2.5$ s **A1**

(c) $s = \displaystyle\int_0^{2.5} \frac{30}{1+0.6t}\,\mathrm{d}t$ **M1**

$= \Big[50\ln(1+0.6t)\Big]_0^{2.5} = 50\ln 2.5$ **(A1)**

$= 45.8$ m $\quad (45.8145\ldots)$ **A1**

The GDC value of the integral is acceptable with the integral shown (M1A1A1). Wrong method: assuming constant deceleration, $\frac{30+12}{2} \times 2.5 = 52.5$*: M0A0A0. FT from (b).*

(d) $v\dfrac{\mathrm{d}v}{\mathrm{d}s} = -0.02v^2, \quad \displaystyle\int \frac{1}{v}\,\mathrm{d}v = \int -0.02\,\mathrm{d}s$ **M1**

$\ln v = -0.02s + \ln 30$ **A1**

$v = 30e^{-0.02s}$ **A1**

A1 for the integrated equation with the constant found from $v = 30$ *at* $s = 0$*.*

(e) $v = 30e^{-0.02(100)} = 30e^{-2}$ **M1**

$v = 4.06$ m s^{-1} $\quad (4.06005\ldots)$ **A1**

Wrong method: $t = 100$ *in* $v(t)$ *gives* 0.492*: M0A0. FT from (d).*

5. [Maximum mark: 13]

(a) $f'(0) = 0 + 2(1) = 2$ **A1**

$y'' = 1 + 2y'$ **M1**

$f''(0) = 1 + 2(2) = 5$ **A1**

$y''' = 2y'', \quad f'''(0) = 2(5) = 10$ **A1**

M1 for differentiating the equation implicitly. Wrong method: differentiating x as 0 ($y'' = 2y'$, giving 4 and 8): M0A0A0.

(b) $y \approx 1 + 2x + \dfrac{5}{2!}x^2 + \dfrac{10}{3!}x^3$ **M1**

$y \approx 1 + 2x + \dfrac{5}{2}x^2 + \dfrac{5}{3}x^3$ **A1**

FT from (a). Dividing by n instead of $n!$ shows only in the x^3 term ($\frac{10}{3}x^3$): A0.

(c) $\dfrac{\mathrm{d}y}{\mathrm{d}x} - 2y = x, \quad$ integrating factor $e^{\int -2\,\mathrm{d}x}$ **M1**

$= e^{-2x}$ **A1**

$\dfrac{\mathrm{d}}{\mathrm{d}x}\left(ye^{-2x}\right) = xe^{-2x}, \quad \displaystyle\int xe^{-2x}\,\mathrm{d}x = -\frac{x}{2}e^{-2x} + \int \frac{1}{2}e^{-2x}\,\mathrm{d}x$ **M1**

$ye^{-2x} = -\dfrac{x}{2}e^{-2x} - \dfrac{1}{4}e^{-2x} + C$ **A1**

$y = 1$ at $x = 0$: $\; C = \dfrac{5}{4}, \quad y = \dfrac{5}{4}e^{2x} - \dfrac{x}{2} - \dfrac{1}{4}$ **A1**

M1 for an integrating factor from the equation in the form $y' + P(x)y = Q(x)$. Second M1 for integration by parts. The final A1 needs the constant found. Wrong methods: integrating factor e^{2x} (sign of P): M1A0, later marks FT; a sign slip in the parts gives $y = \frac{3}{4}e^{2x} - \frac{x}{2} + \frac{1}{4}$: A0 for the integral and the final A1.

(d) $f(0.2) \approx 1 + 0.4 + 0.1 + 0.01333\ldots = 1.51 \quad (1.51333\ldots)$ **A1**

exact: $f(0.2) = \frac{5}{4}e^{0.4} - 0.1 - 0.25 = 1.51478\ldots$

$\text{percentage error} = \dfrac{|1.51333\ldots - 1.51478\ldots|}{1.51478\ldots} \times 100 = 0.0956\%$ **A1**

Accept any value consistent with the candidate's own series and exact solution, or with a GDC value of the true solution. FT from (b) and (c).